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Emerging fungal infections in solid organ transplant recipients: Guidelines of the American Society of Transplantation Infectious Diseases Community of Practice

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[Correction added on May 24, 2019, after first online publication: The second author, Deepali Kumar, was removed from the byline by agreement of all authors. Her affiliation was also removed, and the affiliations renumbered accordingly.]

Abstract

These updated AST-IDCOP guidelines review the epidemiology, diagnosis, and management of emerging fungi after organ transplantation. Infections due to numerous generally innocuous fungi are increasingly recognized in solid organ transplant (SOT) recipients, comprising about 7%-10% of fungal infections in this setting. Such infections are collectively referred to as emerging fungal infections and include Mucormycetes, Fusarium, Scedosporium, and dematiaceous fungi among others. The causative organisms are diverse in their pathophysiology, uncommon in the clinical setting, have evolving nomenclature, and are often resistant to multiple commonly used antifungal agents. In recent years significant advances have been made in understanding of the epidemiology of these emerging fungal infections, with improved diagnosis and expanded treatment options. Still, treatment guidelines are generally informed by and limited to experience from cohorts of patients with hematological malignancies and/or solid and stem cell transplants. While multicenter randomized controlled trials are not feasible for these uncommon infections in SOT recipients, collaborative prospective studies can be valuable in providing information on the epidemiology, clinical manifestations, treatment strategies, and outcomes associated with the more commonly encountered infections.

KEYWORDS
antifungal therapy, fusarium, scedosporium, zygomyces

1 | INTRODUCTION

Infections due to a variety of generally innocuous fungi are increasingly recognized as a problem in solid organ transplant (SOT) recipients.[1-5] These fungi are diverse, uncommon, have evolving nomenclature, and are often resistant to multiple commonly used antifungal agents. In recent years there has been an increased understanding of the epidemiology of such emerging fungal infections, improved diagnostic capabilities, and expanded treatment options. Still, experience to guide treatment is generally limited. Hence, all of the treatment recommendations in these guidelines are derived from small case series, anecdotal experiences, and joint center reviews and are summarized in Table 2 (moderate- to low- quality evidence). Data gleaned from non-SOT populations, such as patients with hematological malignancies and/or hematopoietic stem cell transplant (HSCT), further inform decisions regarding these infections.[6-11]

The causative organisms include filamentous fungi such as the mucormycetes, *Fusarium*, *Scedosporium*, and *Lomentospora prolificans* (formerly *Scedosporium prolificans*), yeast-like organisms such as *Trichosporon*, *Malassezia*, and *Rhodotorula*, the dimorphic

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Emergomyces species, *Talaromyces marneffei* (formerly *Penicillium marneffei*), and *Sporothrix*, and the dematiaceous fungi (a collective term referring to a variety of darkly pigmented fungi).[6,12-18]

Diseases caused by these diverse fungi are collectively known as "emerging" or "rare" fungal infections. A range of newly described *Candida* and *Aspergillus* species can also be termed emerging fungi, but will not be included in this article, as they are more appropriately addressed in those respective guidelines. Since data are limited, distinctions between adult and pediatric patients are not addressed in these guidelines.

2 | EPIDEMIOLOGY AND RISK FACTORS

Overall, emerging fungal infections comprise a small minority (~7%-10%) of fungal infections in SOT recipients.[12,13] Development of infection requires that a susceptible host comes in contact with the causative fungus and that the fungus survives and thrives within the patient.

2.1 | Transplant type and timing of infection

All transplant patients are at risk and infections can occur at all times along the post-transplant continuum. By sheer numbers, most emerging fungal infections occur in kidney transplant recipients.[19,20] However, on a per patient basis, lung and liver transplant recipients are at highest risk. For example, the incidence of mucormycosis in the first year after transplant is 0.07%, but is more than twice that for lung and liver recipients.[21] Infections develop at median onset after transplant of about 10 months for mucormycosis, a year for fusariosis, and 15 months for non-Aspergillus molds.[17,22] However, there is much variability, as such infections tend to occur earlier in liver and heart transplant recipients than in other groups. Furthermore, earlier presentation in the liver transplant recipients is associated with disseminated disease.[17,20] With regard to specific infections, mucormycosis tends to be seen early after transplant. In liver recipients, this can be in the first month after transplant, while in others at 3-6 months or later.[23] Infections due to dematiaceous molds and *Fusarium* on the other hand, often develop years after transplant. Pretransplant colonization is also a factor. For example, scedosporiosis can develop at one month after transplant in those colonized before transplantation, but develops at 6 months or later in others.[24]

2.2 | Fungal exposure

Exposure to emerging fungi is generally due to direct cutaneous contact and/or inhalation from environmental sources such as soil, vegetation, water, sewage, or air. The latter is important, and many people do not have a specific known exposure to fungal spores, but still develop invasive infections. While these organisms are encountered worldwide, rates of infection can vary by geographic locale and intensity of environmental exposure.[25]

Most infections start in the respiratory tract or skin. The ability to grow well at body temperature is a key determinant as to whether deeper tissues can be infected. Thermotolerant organisms (eg, mucormycetes, *Fusarium*, *Scedosporium*, and *Lomentospora*) can invade internal organs, while many other potentially pathogenic fungi (eg, dermatophytes, some dark walled fungi) survive best in colder temperature and hence only infect the cooler cutaneous structures.

The size of airborne fungal propagules often dictates modes of transmission and clinical manifestations. For example, *Aspergillus fumigatus* conidia are ideally suited by size for deposition in the alveoli, and tend to cause pneumonitis. Pulmonary disease is also the most common manifestation of infection with the mucormycetes and *Fusarium*. However, their larger conidia can also get trapped in the upper airways and sinuses, which sometimes favors extra-pulmonary presentations such as sinusitis or infections at sites of direct inoculation. Once the organisms become invasive, some develop smaller morphologies (eg, *Fusarium*, *Scedosporium*, and yeasts), which provide enhanced ability to cause invasive and potentially disseminated disease.

2.3 | Host factors

Major host risk factors for emerging fungal infections include prolonged and profound immunodeficiency, breaks in skin integrity, and chronic respiratory disease (eg, cystic fibrosis and bronchiectasis).[7,9,11,26] In the latter, underlying pulmonary architectural distortion and mucosal defects predispose patients to chronic colonization and infection with *Scedosporium* and *Lomentospora*, mucormycetes, *Purpureocillium lilacinus* (formerly *P lilacinus*), and dematiaceous molds (eg, *Exophiala dermatitidis*) before and after lung transplantation.[6,12,27] Such patients can also be colonized with organisms of lower pathogenic potential (eg, *Paecilomyces variotii*, *Penicillium*, and *Cladosporium*) that in the appropriate clinical scenario can cause invasive disease.

Exposure to antifungal agents may select for less common fungi and contribute to shifts in incidence of infection with the emerging fungi. For example, voriconazole usage has been associated with mucormycosis in HSCT recipients in some medical centers.[28,29] Receipt of inhaled amphotericin B (AmB), may select out for organisms resistant to that drug such as *Scedosporium species*, *L prolificans*, and *P lilacinus*.[31] Much less commonly, infection may be transmitted during the transplantation process via contamination of the preservation fluid or from the organ itself. Transmission of *Scedosporium* and the mucormycetes has been reported in such circumstances and is associated with high rates of graft loss and mortality.[32,33] Donors exposed to contaminated water such as near drowning may be at particular risk for transmitting mucormycetes.

2.4 | Epidemiology of specific fungi

2.4.1 | Mucormycetes

Invasive mucormycosis is a potentially devastating infection with an overall case fatality rate of 40%-50%.[34,35] Important mucormycetes include *Rhizopus*, *Mucor*, *Lichthemia*, *Syncephalastrum*, *Saksenaea*,

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Cunninghamella, *Rhizomucor*, and *Apophysomyces*.[19] Mucormycetes spores are ubiquitous in the air, but particularly associated with natural composts, tropical soils, and other heated materials, poorly maintained heating ducts and attached humidifier structures, soils of potted plants, and indoor composts.[37]

Like invasive aspergillosis, mucormycosis may cause hemorrhagic necrosis, vascular thrombosis, and tissue infarction and can extend locally to infect adjacent structures or disseminate to other sites. Traditional risk factors include uncontrolled diabetes mellitus, corticosteroids, and neutropenia.[23] Other risk factors uniquely described in SOT recipients include renal failure and prior voriconazole and/or caspofungin use.[38] Cases typically develop within 3-6 months of transplant but may occur much later except in liver transplant recipients where disease frequently occurs in the first month after transplant.[38]

2.4.2 | Scedosporium and lomentospora

Approximately 25% of non-*Aspergillus* mold infections are caused by *Scedosporium* and *L prolificans*, which represent 1% of all fungal infections in SOT recipients.[18,21] The genus *Scedosporium* includes several potentially pathogenic species including *Scedosporium apiospermum* (and the related *Pseudallescheria boydii*) and *Scedosporium aurantiacum*.[39] *Lomentospora prolificans*, a dark walled fungus was previously grouped among the *Scedosporium*, has now been classified into its own genus.

Scedosporium and *Lomentospora* are soil saprophytes commonly found in temperate climates (especially in Australia), are associated with nutrient-rich substrates such as soil and manure, and are seen in industrial areas, city parks, and playgrounds, and effluents from wastewater treatment plants or agricultural lands.[40] Acquisition of *Scedosporium* occurs via direct or inhalational contact with contaminated water, soil, or from unknown sources in many cases.[41]

Risk factors for infection include pretransplant colonization (frequently encountered in cystic fibrosis), prior receipt of AmB (to which *Scedosporium* species are generally resistant), and enhanced immunosuppression with treatment for organ rejection.[42,43] The most common infecting species is *S apiospermum*, and the majority of infections are in lung transplant recipients.[18] Median time to infection is approximately 3-4 months after transplantation but can vary, particularly in lung transplant recipients.[2,44,45]

2.4.3 | Fusarium

Fusariosis accounts for approximately 13% of non-*Aspergillus* mold infections and 0.6% of all fungal infections in SOT recipients.[18,21] *Fusarium solani*, *Fusarium proliferatum*, *Fusarium oxysporum*, *Fusarium moniliforme*, and *Fusarium sacchari* have been implicated as causative agents of infections.[47,48] *Fusarium* is acquired from environmental sources, air, tap water, sinks, and showerheads.[49] The portal of entry is usually the skin or respiratory tract. Risk factors include persistent neutropenia, profound T-cell depletion and previous fungal infections. In SOT recipients, fusariosis is generally manifested as localized cutaneous infection with the notable exception of lung transplantation, where pneumonia is the predominant presentation.[22]

2.4.4 | Dematiaceous fungi

Infections due to dematiaceous (dark walled) fungi are termed phaeohyphomycosis and comprise about 2.5% of invasive fungal infections in SOT patients.[50] Of the more than 100 different species in this group, the most important ones for SOT recipients are *Alternaria*, *Exophiala*, and *Cladophialophora*. Others occasionally encountered include *Curvularia*, *Exserohilum*, *Ochroconis*, *Rhinocladiella mackenziei*, *Scopulariopsis*, and *Wangiella*.[51] They are generally found in soil and in association with plants. Distribution is worldwide, although environmental distribution may vary. For example, *R mackenziei* infections are primarily seen in patients from the Middle East.

The nomenclature is constantly evolving. For example, *Wangiella dermatitidis* may be referred to as *E dermatitidis* in some articles, and *L prolificans*, is typically categorized with the Scedosporium species, with which it was previously classified.

In SOT recipients, these fungi can cause localized cutaneous and subcutaneous infections (~50 of cases), infections of the sinuses, eyes, lungs, bones and joints, and disseminated diseases such as infection at multiple sites, fungemia, and central nervous system involvement.[50,52] *Alternaria* is the most common infecting species in transplant recipients and typically causes superficial and/or deep localized disease. *Cladophialophora bantiana* and *R mackenziei* on the other hand are associated with brain abscesses.

Presence of dematiaceous fungi in respiratory tract or sinus culture does not necessarily indicate invasive disease. For example, *Cladosporium* is rarely pathogenic and *Curvularia* and *Bipolaris* are frequently associated with allergic rather than invasive sinusitis. However, identification of a dematiaceous fungus from a clinical specimen should not be carelessly dismissed as such fungi are increasingly recognized as important causes of infections in SOT recipients.

2.5 | Diagnosis

Colonization with one of the emerging fungi may occur in the recipient before or after transplantation. The presence of an emerging fungal species in cultures obtained from non-sterile sources does not necessarily indicate infection. This is a particularly relevant issue in lung transplant recipients in whom a variety of emerging fungi including the dematiaceous molds (eg, *Exophiala* and *Cladosporium*), *Scedosporium*, Basidiomycetes, mucormycetes, *Fusarium*, *Paecilomyces*, and *Penicillium* can colonize the respiratory tract before or after transplantation.[53,54] Frequency of colonization with specific fungi is dependent on the underlying condition and geographic locale. Evidence demonstrating a direct correlation to post-transplant fungal infection has varied among transplant centers.[47,57,58] Likewise, pulmonary colonization after lung transplantation does not necessarily lead to invasive infection, even in the absence of antifungal therapy. In the clinical setting, distinguishing

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colonization from invasive infection can be challenging and often requires tissue examination. In renal transplant recipients, isolation of *Trichosporon* species in urine cultures is usually a benign finding and rarely associated with invasive or deep-seated infection.[61,62]

Infection due to emerging molds and yeasts can be difficult to diagnose. Clinical signs and symptoms are non-specific and indistinguishable from more common fungal infections. A comprehensive diagnostic approach that includes invasive procedures (eg, bronchoalveolar lavage, biopsies), careful specimen collection and processing, utilization of specific culture media, and select histological staining techniques is usually necessary for establishing the diagnosis.[13] Histopathologic analysis of biopsy specimens that demonstrate septate hyphae on hematoxylin and eosin (H&E) staining may be seen with *Aspergillus*, *Fusarium*, and *Scedosporium*.[10] In contrast, mucormycetes typically appear as broad, nonseptate/pauciseptate, ribbonlike hyphae by H&E staining. Fontana-Masson staining can be valuable for identifying dematiaceous fungi in tissue. Delayed or incorrect identification may lead to the initiation of incorrect treatment and results in further tissue destruction and/or dissemination of disease. Close communication between the transplant team and the mycology and pathology laboratory is essential. Final identification and susceptibly testing frequently requires referral to a reference laboratory. Once the fungus has been identified, distinguishing colonization from active infection is a potentially challenging but essential component of the pre- and post-transplant evaluation.[3,44] Radiographic studies can demonstrate pathologic changes in tissue, particularly the lung, and help distinguish disease from colonization; however, imaging findings are not specific.

Non-culture-based diagnostic assays are increasingly employed to aid in diagnosis of a range of fungal infections. Serum and BAL galactomannan assays are predominantly helpful for aspergillosis, but might be positive in cases of infection with *Alternaria*, *Penicillium*, *Paecilomyces*, *Fusarium*, and rarely with mucormycetes.[63,64] Assays that detect beta-glucan (generally in blood, but also in cerebrospinal fluid) may be effective at early detection of a range of fungi, including *Fusarium* and *Trichosporon*. However, the assay lacks specificity for any fungal pathogen and does not detect mucormycetes.[69,70] Utilization of nucleic acid amplification tests, such as fungal sequencing targeting the ribosomal ITS2 and D2 regions, can be highly sensitive and specific for identification of fungi. These perform best when a sample is obtained via open biopsy and the pre-test probability for an invasive fungal infection is particularly high.[71]

For organisms that grow on culture, matrix-assisted laser desorption ionization-time-of-flight mass spectrometry (MALDI-ToF MS) is increasingly incorporated into clinical microbiology workflows to help identify emerging fungal organisms down to the species level. As MALDI-ToF databases are further developed the utility of this technology for rapidly identifying uncommon and emerging fungal is expected to increase.[72]

3 | CLINICAL MANIFESTATIONS

The clinical manifestations and diagnosis of the emerging fungal infections are summarized in Table 1.

4 | TREATMENT

4.1 | Overview/general principles

Treatment of emerging fungal infections continues to be a challenge. Many of the antifungal agents required for treatment are expensive, have the potential for drug interactions and/or substantial toxicity, and need to be taken for prolonged periods. Given these limitations, we recommend the following approach:

1. Therapy with an agent that has proven activity against the fungus should be administered as early as possible (Strong, Low).
2. Immunosuppression should be reduced when clinically feasible. To date, immune reconstitution inflammatory syndrome has not been described with emerging or rare fungal pathogens (Strong, Low).
3. Surgical debridement of the affected areas should be performed whenever feasible and repeated as needed (Strong, Low).
4. Antifungal therapy should be adjusted based on susceptibility testing at a reference laboratory. Although clinically validated antifungal susceptibility breakpoints are lacking, it is reasonable for therapy to be guided by general antifungal susceptibility patterns (Weak, Low).
5. Clinicians should closely monitor for renal toxicity with AmB products. If using such agents, treatment should generally be with a lipid formulation of AmB (LF-AmB; ie, liposomal or lipid complex formulations) as these have less toxicity (Strong, Low).
6. Clinicians should closely monitor for Q-T interval prolongation, drug interactions, hepatotoxicity, and neuropsychiatric side effects with azoles.[73,74]
7. Therapeutic drug monitoring of voriconazole should be considered to guide dose adjustments, especially in patients receiving higher doses or corticosteroids, although data for emerging fungal infections are lacking. Target voriconazole trough levels between 1.5-4.5 mcg/mL are associated with the optimal balance of maximizing efficacy and minimizing toxicity.[75] Based on very limited data in the prophylactic, setting the target posaconazole trough levels should be at least 0.7 mcg/mL.[76,77] Routine monitoring of posaconazole levels may be unnecessary. We recommend checking serum trough levels when using the liquid formulation, in special populations (eg, patients with very high BMI, questionable drug absorption and those with cystic fibrosis) and when clinical response is suboptimal. It is unclear whether therapeutic drug levels are useful for isavuconazole.
8. Adjuvant therapy with interferon-gamma and/or granulocyte-macrophage colony stimulating factor may be considered with caution in cases refractory to standard antifungal therapy.[78,79] (Weak, Very Low)

The recommended treatment of mucormycosis, *Fusarium*, *Scedosporium* and other emerging fungal infections is summarized in

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TABLE 1 Clinical manifestations and diagnosis of emerging fungal infections

Fungal pathogen (References)	Clinical manifestations	Diagnosis[a]
Mucormycetes [19,23]	▪ Pulmonary disease (most common) ▪ Rhino-orbital-cerebral (associated with diabetic ketoacidosis) ▪ Disseminated disease (most common in liver transplant recipients) ▪ Primary cutaneous disease (frequently at sites of trauma or medical/surgical interventions) ▪ Bronchial anastomosis infections in lung transplant recipients ▪ Gastrointestinal	▪ Broad, ribbonlike, nonseptate/pauciseptate hyphae in tissue with calcofluor, PAS, or GMS silver staining ▪ Growth in culture usually within 24-48 h ▪ Cultures frequently negative despite visualization on stains
Fusarium [22,98]	▪ Superficial and deep cutaneous disease (most common in solid organ transplant) ▪ Pulmonary disease ▪ Disseminated (usually involves lung and skin) ▪ Fungemia ▪ Sinusitis ▪ Osteomyelitis ▪ Keratitis (especially in agricultural areas and with contact lens wear) ▪ Peritonitis ▪ Endocarditis	▪ Blood cultures may be positive ▪ Histopathologic appearance is similar to *Aspergillus*
Scedosporium and Lomentospora [24]	▪ Lung infection ▪ Disseminated infection ▪ CNS infection ▪ Skin (less common) ▪ Bone and joint infection ▪ Ocular infection ▪ Hepatosplenic infection ▪ Peritonitis ▪ Endovascular infection	▪ Histopathologic appearance is similar to *Aspergillus* ▪ Blood cultures may be positive ▪ Lomentospora reacts with Fontana-Masson stain
Paecilomyces and Purpureocillium lilacinus (formerly *Paecilomyces lilacinus*) [99,100]	• Insidious cutaneous and subcutaneous infections: o Cellulitis in areas of trauma o Erythematous, violaceous, or crusted ulcerations, plaques, nodules, and papulopustular lesions o Sporotrichoid pattern • Sternal wound infection in a lung transplant recipient with pretransplant respiratory colonization • Other manifestations: sinus disease, osteomyelitis, keratitis, endophthalmitis, and disseminated infection	▪ Reference laboratory should be consulted for confirmation ▪ Irregular septate hyphae in tissue with PAS or GMS silver stain can be confused with other molds. ▪ Suppurative and/or granulomatous inflammation in tissue
Trichoderma (Hyalohyphomycosis) [101]	▪ Perihepatic abscesses in liver transplant recipients ▪ Pleuropulmonary disease in lung transplant recipients ▪ Disseminated infection involving brain in kidney transplant recipients	▪ Fine hyaline septate hyphae in tissues with positive cultures
Scopulariopsis	▪ Disseminated infection with cutaneous, pleuropulmonary, cardiac, and brain involvement with nearly universal mortality [102,103]	▪ Branched and septate hyphae with GMS silver stain
Acremonium [106]	▪ Mycetoma	
Dematiaceous fungi [52,107,108]	▪ Subcutaneous nodules and less commonly as skin abscesses, pustular lesions, or purulent ulcerations	▪ Septate hyphae with GMS silver stain ▪ Fontana-Masson staining highlights the presence of melanin in the dematiaceous fungi
Chromoblastomycosis	▪ Chronic cutaneous disease most frequently in the tropical and subtropical areas	▪ Pigmented sclerotic bodies with hematoxylin and eosin stain ▪ Septate hyphae with lactophenol alanine blue stain
Trichosporon [110,111]	▪ Typically catheter associated ▪ Disseminated infection	▪ Blood cultures are typically positive ▪ Budding yeast in tissue

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TABLE 1 (Continued)

Fungal pathogen (References)	Clinical manifestations	Diagnosis[a]
Malassezia [114,115]	▪ Pityriasis or tinea versicolor, folliculitis ▪ Onychomycosis ▪ Groin abscess	▪ KOH preparation and/or culture.
Rhodotorula [116,117]	▪ Peritonitis in a liver transplant recipient ▪ Fungemia in a liver-kidney transplant recipient	▪ Budding yeast
Talaromyces marneffei (formerly *Penicillium marneffei*) [118,119]	▪ Endemic in Southeast Asia, southern China, Taiwan, and Hong Kong ▪ Disseminated infections	▪ Dimorphic fungus ▪ Produces a red dye in culture ▪ Yeast forms can be elongated and have well-demarcated septa
Paracoccidioides [123,124]	▪ Endemic to Latin America ▪ Pulmonary involvement with or without disseminated disease	▪ Dimorphic fungus ▪ Yeast forms bud off forming the characteristic "pilot's wheel" morphology
Sporothrix[126,127]	▪ Infection is primarily initiated by trauma to the skin resulting in cutaneous infections marked by suppurative and granulomatous nodules that spread along lymphatic channels ▪ Disseminated infection ▪ Pulmonary infection in a heart transplant patient	▪ Dimorphic fungus ▪ Culture is gold standard as smear often negative for organisms

Nucleic acid amplification tests (eg, fungal sequencing targeting the ribosomal ITS2 and D2 regions) are in development for diagnosis of multiple uncommon and emerging fungi. Matrix-assisted laser desorption ionization-time-of-flight mass spectrometry (MALDI-ToF MS) is increasingly utilized for aid in identification of organisms in culture down to species level. PAS, periodic acid schiff; GMS, grocott's methenamine silver.
[a]When feasible, we recommend speciation of the organism and susceptibility testing in cases of invasive infection.

Table 2. Target length of therapy should be individualized based on response to therapy and safety considerations. As a general rule and if feasible, antifungal therapy should be continued at least until all clinical and radiographic signs of infection have resolved.

4.2 | Treatment of specific fungi

4.2.1 | Mucormycetes

The angiotropic nature of mucormycosis leads to tissue infarction and necrosis. Consequently, surgical resection or debridement is nearly always required unless infection is limited to the lungs.[80] Even then, for bronchial anastomotic infections in lung transplant recipients, bronchoscopic or surgical debridement is essential.[81] Medical therapy alone can be attempted in patients with pulmonary mucormycosis unless there is extensive necrosis or disease threatening major vascular structures. LF-AmB is the treatment of choice. In a study of cancer patients with mucormycosis, early treatment (within 5 days of diagnosis) significantly improved outcome.[82]

Posaconazole may be considered for maintenance therapy once clinical stability has been achieved (Weak, Low). Evidence is inconclusive regarding benefit of combination therapy with LF-AmB and posaconazole, either as frontline or as salvage treatment in those failing to respond to initial treatment.[83-85] More recently, isavuconazonium sulfate (prodrug of isavuconazole) was evaluated in an open-label study of patients with mucormycosis. Mortality at day 42 was 37.8%, approximately the same seen in patients receiving AmB.[86] Isavuconazole is a reasonable second-line option for mucormycosis (Weak, Moderate).[86,87]

4.2.2 | Scedosporium and Lomentospora

Response to therapy is highly dependent on site of infection, extent of dissemination, and host factors.[46] Outcomes are better when the infection is localized to the skin or lung and substantially worse with disseminated disease. Surgical excision is typically required. Outcomes tend to be better with *S apiospermum* infection, which may be related to better response to antifungal agents.[88,89] In vitro, voriconazole has the most potent activity against this species. The echinocandins are also active. Other available drugs (eg, AmB, 5-flucytosine, and terbinafine) are essentially inactive against *S apiospermum*.[89] Medical management of *L prolificans* (formerly *S prolificans*) is extremely challenging, and surgery is typically required to control infection. This species is resistant to virtually all of the available antifungal agents.[89] Based on animal studies, combination therapy including an echinocandin and either AmB or voriconazole may be effective.[90] In vitro and anecdotal reports suggest that combining voriconazole with terbinafine may be effective.[91,92] However, while terbinafine has some in vitro synergy, this agent is keratinophilic and hence does not penetrate well into deeper tissues.

4.2.3 | Fusarium

Surgical resection alone may be effective for limited cutaneous or sinus disease. Antifungal therapy is recommended for deeper sites of infection (eg, lungs) or disseminated disease.[93] Voriconazole is generally first-line.[94] Antifungal susceptibility can vary by species and in vitro testing should guide choice of antifungal therapy. In

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TABLE 2 Recommended treatment of emerging and rare fungal infections in solid organ transplant recipients

Fungal pathogen (references)	Initial therapy	Alternative therapy
Mucormycetes	Surgical excision/debridement is recommended for all infections outside of lungs Induction therapy: • LF-AmB is the treatment of choice (Strong, Moderate) • Combination of an echinocandin + LF-AmB may be considered based on animal studies and retrospective reports [129,130] (Weak, Very Low) • AmB deoxycholate was historically the drug of choice but is associated with substantial nephrotoxicity and generally avoided in the current era [80,132,133] (Strong, Moderate) Maintenance therapy: • Posaconazole or isavuconazonium[a] sulfate (Weak, Moderate)	Induction therapy: • Posaconazole or isavuconazonium sulfate[a] for patients intolerant to or failing LF-AmB [86,134,135] (Weak, moderate) • Combination of posaconazole + LF-AmB may be considered in refractory infections, but data are conflicting (Weak, Very Low) Maintenance therapy: • LF-AmB in patients who are clinically unstable or unable to tolerate oral intake
Fusarium [137]	Surgical excision or debridement is recommended whenever feasible • Voriconazole (Weak, Low)	• Combination therapy (LF-AmB + voriconazole[b]) may be considered pending final identification and susceptibility data. Goal is to provide coverage for organisms resistant to one or the other of these agents. (Weak, Low) • Posaconazole (Weak, Low) • *Fusarium solani* and *Fusarium verticilloides* may be resistant to voriconazole and require LF-AmB (Weak, Low)
Scedosporium	Surgical excision or debridement is recommended whenever feasible • Voriconazole[b] [88,89] (Weak, Low)	• Combination of an echinocandin and voriconazole may be considered (Weak, Very Low)
Lomentospora prolificans	Surgical debridement should be considered primary therapy (resistant to virtually all of the available antifungal agents) [89] • Echinocandin + LF-AmB or voriconazole [90] (Weak, Low) • Voriconazole + terbinafine[d] [91,92] (Weak, Low)	
Purpureocillium [99,100,138]	Surgical excision or debridement is recommended whenever feasible and may be sufficient for isolated cutaneous disease • Voriconazole or posaconazole for more extensive disease (Weak, Low)	
Trichoderma [101,139]	Surgical excision or debridement is recommended whenever feasible • Antifungal susceptibility is necessary to guide therapy • Combination therapy LF-AmB + voriconazole or posaconazole may be considered pending susceptibility data (Weak, Low)	
Scopulariopsis and *Microascus* [140,141]	Surgical excision or debridement is recommended whenever feasible • Antifungal susceptibility is necessary to guide therapy • Combination therapeutic options based on in vitro synergy data (Weak, Low) o Terbinafine + voriconazole or posaconazole o Caspofungin + voriconazole or posaconazole	
Acremonium [142]	Surgical excision or debridement is recommended	
Phaeohyphomycosis *Exophiala* [52,107,109,143,144]	Surgical excision or debridement is recommended whenever feasible and may be sufficient for isolated cutaneous disease • Antifungal susceptibility is necessary to guide therapy • Voriconazole, posaconazole, or itraconazole are first-line agents (Weak, Low)	• Echinocandins may be considered based on in vitro data • Potential combination therapeutic options based on in vitro synergy data • LF-AmB + flucytosine (Weak, Very Low) • Itraconazole + flucytosine (Weak, Very Low)

(Continues)

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TABLE 2 (Continued)

Fungal pathogen (references)	Initial therapy	Alternative therapy
Chromoblastomycosis	• Surgical excision or debridement is recommended whenever feasible and may be sufficient for isolated cutaneous disease • Antifungal susceptibility is necessary to guide therapy • Voriconazole, posaconazole or itraconazole are first-line agents (Weak, Low)	• Echinocandins may be considered based on in vitro data (Weak, Very Low)
Trichosporon	• Identification in urine of kidney transplant recipients generally does not require treatment • Antifungal susceptibility is necessary to guide therapy • Azoles are recommended first-line agents (Strong, Low)	• LF-AmB may be considered if susceptibility is confirmed (Weak, Low)
Malassezia	• Topical preparation of clotrimazole 1% and selenium sulfide lotion or oral fluconazole for superficial infections • Identification to species level and antifungal susceptibility is necessary to guide therapy • Catheter removal and LF-AmB for disseminated infection (Weak, Low)	• Itraconazole, posaconazole, voriconazole (Weak, Low)
Rhodotorula [140]	• LF-AmB or posaconazole are treatments of choice based on available susceptibility data (Weak, Low)	
Penicillium [145]	• Terbinafine and the echinocandins show the best in vitro activity (Weak, Low)	
Talaromyces marneffei (formerly *Penicillium marneffei*)	• Induction treatment with LF-AmB followed by maintenance treatment with itraconazole is recommended (Strong, Moderate)	• Itraconazole (Weak, Low) • Voriconazole (Weak, Low)
Paracoccidioides [146]	• Induction treatment with LF-AmB for severe disease followed by maintenance treatment with itraconazole or voriconazole is recommended (Weak, Low) • Itraconazole may be considered for less severe disease (Weak, Low) • May be prevented with trimethoprim-sulfamethoxazole	• Voriconazole may be considered for less severe disease (Weak, Low) • Trimethoprim-sulfamethoxazole (Weak, Low)
Sporothrix [126,147]	• Induction treatment with LF-AmB for severe disease followed by maintenance treatment with itraconazole (Weak, Low)[c]	• Itraconazole or voriconazole may be considered for less severe disease (Weak, Low)

None of the above treatments are FDA-approved except where noted.
Standard antifungal dosing is recommended: LF-AmB 5 mg/kg daily; AmBd 1-1.5 mg/kg daily; voriconazole 6 mg/kg intravenous q 12 h × 2 loading doses followed by 4 mg/kg intravenous or 200-300 mg orally twice daily; itraconazole 200 mg twice daily; posaconazole delayed release tablets 300 mg daily; caspofungin 70 mg loading dose followed by 50 mg daily; micafungin 100 mg daily; anidulafungin 200 mg loading dose followed by 100 mg daily; isavuconazonium sulfate 372 mg intravenous or oral q 8 h for six doses followed by 372 mg daily).
AmBd, amphotericin B deoxycholate; LF-AmB, lipid formulation of amphotericin B.
[a]Isavuconazonium sulfate is FDA-approved for *Mucorales* infections.
[b]Voriconazole is FDA-approved for *Scedosporium apiospermum* and *Fusarium* infections when intolerant or refractory to other agents.
[c]Amphotericin B deoxycholate is FDA-approved for *Sporothrix* infections.
[d]Terbinafine has some in vitro synergy but the agent is keratinophilic and does not penetrate well into other tissues hence limiting efficacy for invasive disease.

vitro, *Fusarium* species are often resistant to AmB and have a wide range of susceptibilities to voriconazole.[95]

5 | PREVENTION AND PROPHYLAXIS

The most common mechanism for colonization or infection is via environmental exposure. Patients should be instructed to avoid visiting construction sites and poultry farms, manipulating air-conditioning filters, and contact with sewage or decaying material. To reduce the risk of invasive fungal infection due to transmission during the transplantation process, care should be taken in accepting organs from near drowning victims. Organ procurement agencies should report all fungal isolates from a donor to the recipient center. Not all patient with fungal colonization require prophylaxis. Certain colonizing fungi are very rarely pathogenic (eg, *Cladosporium*, *Purpureocillium*, and *Penicillium* species) and their presence generally does not require prophylaxis. By contrast, the mucormycetes and *Scedosporium* have been associated with disseminated infection in highly immunocompromised patients. Prophylaxis may be considered in such patients and in recipients of donor lungs that are colonized with these fungi (Weak, very low)[42,60,96].

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6 | FUTURE DIRECTIONS

There are several antifungal agents in development that show activity against some of the more difficult to treat pathogens such as Scedosporium and *Fusarium*. One novel compound, F901318, is an orotomide and inhibits dihydroorotate dehydrogenase. It has potent in vitro activity against several *Scedosporium* species and *Lomentospora*.[97] Another agent is E1210, which inhibits GPI-dependent anchoring cell wall proteins and has in vitro activity against *Fusarium*.[89] Another prospect is hemofungin, which inhibits ferrochelatase, an enzyme in the heme biosynthetic pathway, and inhibits the growth of *Fusarium*.

While randomized controlled trials are not feasible for these uncommon infections, collaborative prospective studies can be valuable in providing information on the epidemiology, clinical manifestations, treatment strategies, and outcomes associated with mucormycosis, *Fusarium*, and *Scedosporium* in SOT recipients. Transplant professionals should continue to publish case reports and case series of the other emerging fungal infections. Doing so helps define the scope of these uncommon diseases, and when evaluated in the aggregate, can advance knowledge of how to best manage emerging fungal infections.

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